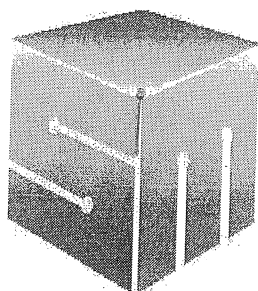




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SM265
Preliminary Datasheet



Silicon Motion, Inc.

SM265

HS MMC Memory Card Controller

Preliminary

Version 1.0

07/04/2006

SM265 DataBook



Silicon Motion®, Inc.

SM265 DataBook

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Revision History

Reversion History

Version	Date	Note
AA0.0	2005/6/9	Preliminary version
AB0.0	2005/7/1	Add 7 x 3.8 package outline and pin definition
0.9	2006/3/10	- Add index - Delete 7x7mm LGA - Add flash interface timing - Add DC/AC parameter
1.0	2006/7/04	- Add MMC4.0/4.1 register table and MMC timing Interface - Add modify the register table to match test flow



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Chapter 1: Overview

General Description

SM265 is the 3rd generation MMC controller of Silicon Motion, Inc. which use 0.18 um process technology. It has higher performance, lower power consumption. SM265 supports both MMC3.3 and MMC4.0/MMC4.1 specification. SM265 has dual-channel high-speed flash interface with RS-ECC, and supports all kinds of NAND type flash which includes MLC, SLC, AG-AND. and 16bit flash module.

Key Features

Features	Benefits
Support MMC specification v3.3 - Current MultiMediaCard version, 20 MHz 1 bit bus. - MMCA compliance program v1.6 verified	Backward compatible with old spec.
Support MMC specification v4.0/4.1 - latest MMC specification, 52MHz, 8 bit data bus. - supports 26/52MHz 1/4/8 bit data bus. - single voltage application ready.	Specifically designed for high speed MMC card.
Support MMC command class: - class 0, 2, 4, 5, 6, 7, 8	Powerful support for SD/MMC common command
Support firmware ISP (in system programming) function, firmware upgradeable.	Firmware update for better compatibility without silicon change
High performance, Low power - Single channel performance up to 20 MB/s - Dual channel performance up to 42 MB/s - Whole card standby current < 150uA - Whole card operation current < 30mA	High performance with lower power consumption
Embedded 4-Symbol Correction ECC function - hardware RS-code ECC, 4 symbol/512byte	Auto error detection and error correction to enhance reliability, Support MLC, AG-AND Flash
Advanced 0.18um process	Advanced technology for better cost structure
Enhanced ESD design	Compliant with ANSI EOS/ESD-S5.1-1998 up to
Flash interface - support single channel 8 bit flash interface - support dual channel 8 bit flash interface - support single channel 16 bit flash interface - each channel support up to 4 flash chips (die/LGA48) - total memory capacity up to 4GBytes	Support high capacity memory card
Build in 1.8V voltage regulator - support single voltage flash interface I/O - regulator support 100mA, 1.8V power supply	To meet low power requirement of mobile devices
Flash memory support - hardware RS-code ECC, 4 symbol/512byte - support Toshiba/SanDisk SLC/MLC NAND type flash - support Samsung/ST/Hynix SLC NAND type flash	High flexible support in various type of Data Flash



- support Samsung/ST/Hynix MLC NAND type flash - support Renesas AG-AND flash - support Micron SLC/MLC NAND type flash - support 512byte/page or 2Kbytes/page architecture - support 4 plane operation	
Package - LGA48 pins - Package: 7mm x 3.8mm - U-type IO pad location - Small die size	

Table 1: Product Benefits

Block Diagram

The following figure is the internal block diagram of SM265.

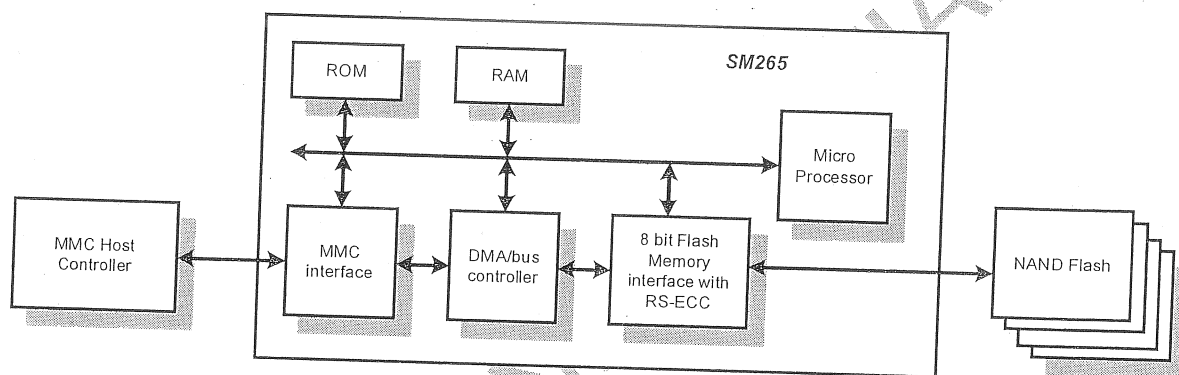


Figure 1 SM265 Internal Block Diagram



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The following figure is the typical application block diagram of SM265.

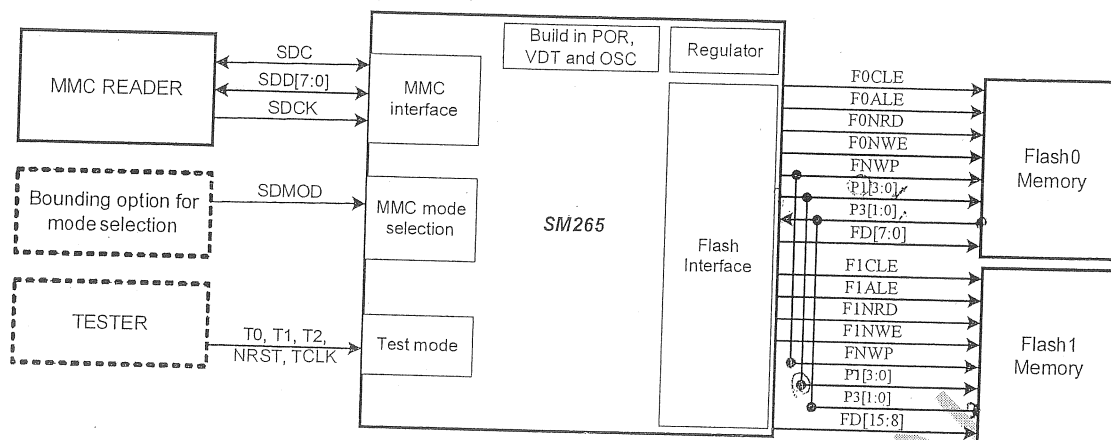


Figure 2 SM265 application block diagram.

Power System of SM265

SM265 is single voltage memory card and the working voltage range is 2.7V to 3.6V. The following figure shows the system block diagram by power connection.

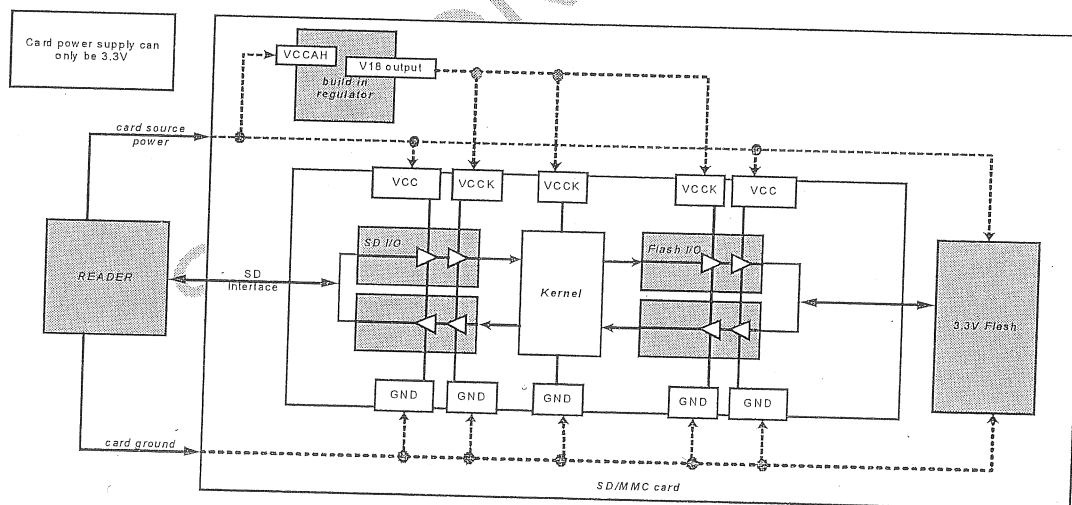


Figure 3 SM265 application block diagram by power connection



Power Supply

SM265 has the following 2 power supply group:

(1) V_{CK} :

Kernel power supply. Input range is 1.65~1.95V. Supplied by the build in voltage regulator whose output pin is V18. The supply current is 100 mA. An 4.7uF capacitance is recommended to connect between pin "V18" and ground to stabilize the current supply.

(2) V_{CC} :

Input/output reference. This group of I/O is supplied direct from the power source of SD/MMC card.

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Chapter 2: Pin Function Description

Flash Interface

Pin Name	Direction	Description
F0D[15:0]	I/O	Flash interface data bus, direct connect to the flash memory.
F0CLE	O	Flash0 interface command latch enable, direct connect to the flash memory.
F0ALE	O	Flash0 interface address latch enable, direct connect to the flash memory.
F0NRD	O	Flash0 interface read strobe control, direct connect to the flash memory.
F0NWE	O	Flash0 interface write strobe control, direct connect to the flash memory.
F1CLE	O	Flash0 interface command latch enable, direct connect to the flash memory.
F1ALE	O	Flash0 interface address latch enable, direct connect to the flash memory.
F1NRD	O	Flash0 interface read strobe control, direct connect to the flash memory.
F1NWE	O	Flash0 interface write strobe control, direct connect to the flash memory.
FNWP	O	Flash interface write protection, direct connect to the flash memory.
P3[1:0]	I	Used as busy signal from flash memory.
P1[3:0]	O	Used as flash chip select signal. Max. 8 flash chips(dual channel) can connect to SM265
VCC	Power in	Power supply for flash group I/O pad.

MMC interface

Pin Name	Direction	Description
SDC	I/O	MMC interface command line
SDD[7:0]	I/O	MMC interface data line
SDCK	I	MMC interface synchronous clock input
VCC	Power in	Power supply for SD group I/O pad.

Mode selection pin

Pin Name	Direction	Description
SDMOD	I	Connect to low for MMC mode.

Test Pin

Pin Name	Direction	Description
T2	I	Test mode selection, tie to <i>low</i> for normal operation.
T1	I	Test mode selection, tie to <i>low</i> for normal operation.
T0	I	Test mode selection, tie to <i>low</i> for normal operation.
NRST, TCLK	I/O	Reserved for CP/FT. tie to <i>high</i> for normal operation.



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Voltage Regulator

Pin Name	Direction	Description
V18	Power out	1.8V / 100 mA power supply. For kernel 1.8V input.
VCCA	Power in	Regulator in. Direct connect to card source power.
GNDA	Ground	Analog ground.

Other Power Pin

Pin Name	Direction	Description
VCKK	Power in	SM265 kernel power supply. 1.8V input.
GND	Ground	Ground.

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Physical Pin Assignment Data Part I : Chip On Board

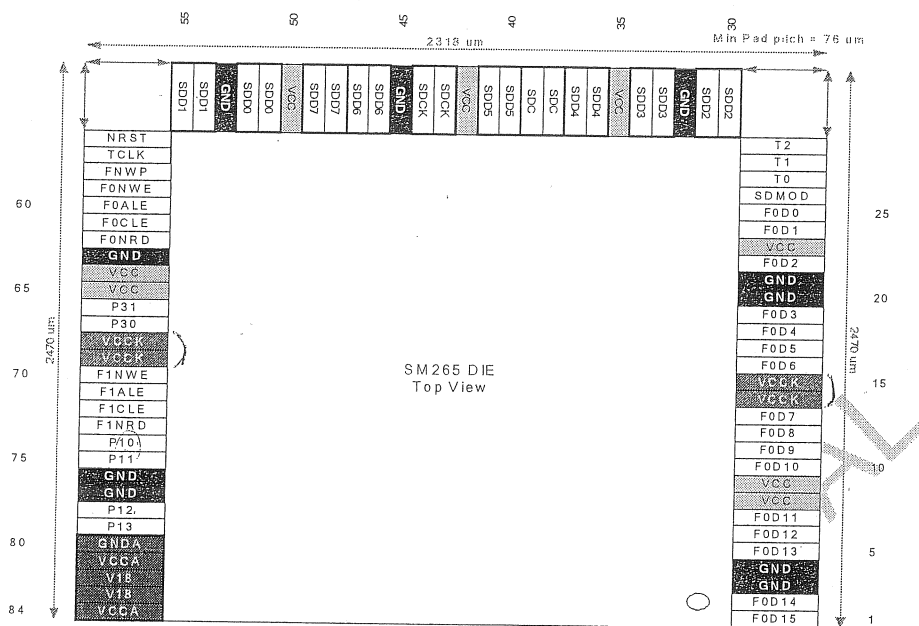


Figure 4 SM265 pin diagram (Top view, die)

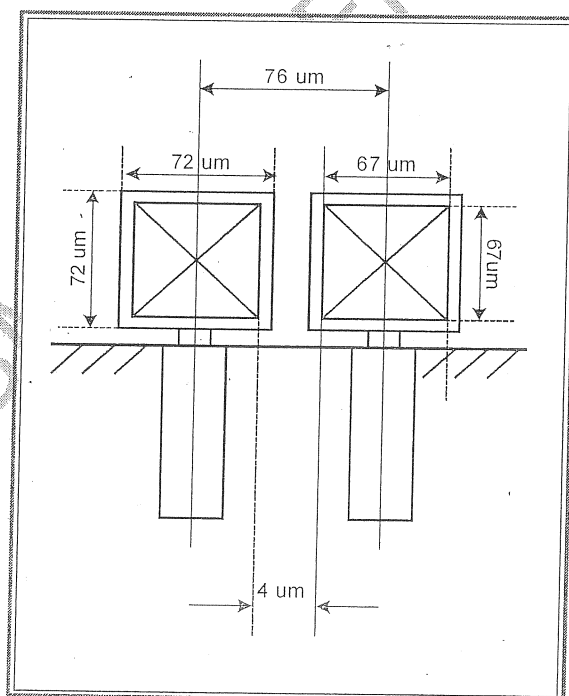


Figure 5 SM265 minimum pad pitch and pad size

SM265 pad size is 72um x 72um,
 Pad opening window is 67um x 67um.
 minimum pad pitch is 76 um.

300 X 300

200k

ROM 12k
 RAM 8k

32kbyte
 Downloading

"NAND" on program.

↓

<Code Swap>



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2318 um x 2470 um

lower left : (0 , 2)

Upper right : (2318.0 , 2469.62)

Pad XY coordinate

Pad No.	Text	X-axis	Y-axis
1	FD[15]	2285.0	114.0
2	F0D[14]	2285.0	190.0
3	GND	2285.0	266.0
4	GND	2285.0	342.0
5	F0D[13]	2285.0	418.0
6	F0D[12]	2285.0	494.0
7	F0D[11]	2285.0	570.0
8	VCC	2285.0	646.0
9	VCC	2285.0	722.0
10	F0D[10]	2285.0	798.0
11	F0D[9]	2285.0	874.0
12	F0D[8]	2285.0	950.0
13	F0D[7]	2285.0	1026.0
14	VCCK	2285.0	1102.0
15	VCCK	2285.0	1178.0
16	F0D[6]	2285.0	1254.0
17	F0D[5]	2285.0	1330.0
18	F0D[4]	2285.0	1406.0
19	F0D[3]	2285.0	1482.0
20	GND	2285.0	1558.0
21	GND	2285.0	1634.0
22	F0D[2]	2285.0	1710.0
23	VCC	2285.0	1786.0
24	F0D[1]	2285.0	1862.0
25	F0D[0]	2285.0	1938.0
26	SDMOD	2285.0	2014.0
27	T0	2285.0	2090.0
28	T1	2285.0	2166.0
29	T2	2285.0	2242.0
30	SDD[2]	2067.835	2441.292
31	SDD[2]	1996.165	2441.292
32	GND	1924.567	2440.045
33	SDD[3]	1852.835	2441.292
34	SDD[3]	1781.165	2441.292
35	VCC	1708.685	2447.127
36	SDD[4]	1637.835	2441.292

Pad No.	Text	X-axis	Y-axis
37	SDD[4]	1566.165	2441.292
38	SDC	1485.835	2441.292
39	SDC	1414.165	2441.292
40	SDD[5]	1333.835	2441.292
41	SDD[5]	1262.165	2441.292
42	VCC	1189.685	2447.127
43	SDCK	1118.835	2441.292
44	SDCK	1047.165	2441.292
45	GND	975.567	2440.045
46	SDD[6]	903.835	2441.292
47	SDD[6]	832.165	2441.292
48	SDD[7]	751.835	2441.292
49	SDD[7]	680.165	2441.292
50	VCC	607.685	2447.127
51	SDD[0]	536.835	2441.292
52	SDD[0]	465.165	2441.292
53	GND	393.567	2440.045
54	SDD[1]	321.835	2441.292
55	SDD[1]	250.165	2441.292
56	NRST	33.0	2242.0
57	TCLK	33.0	2166.0
58	FNWP	33.0	2090.0
59	F0NWE	33.0	2014.0
60	F0ALE	33.0	1938.0
61	F0CLE	33.0	1862.0
62	F0NRD	33.0	1786.0
63	GND	33.0	1710.0
64	VCC	33.0	1634.0
65	VCC	33.0	1558.0
66	P3[1]	33.0	1482.0
67	P3[0]	33.0	1406.0
68	VCCK	33.0	1330.0
69	VCCK	33.0	1254.0
70	F1NWE	33.0	1178.0
71	F1ALE	33.0	1102.0
72	F1CLE	33.0	1026.0

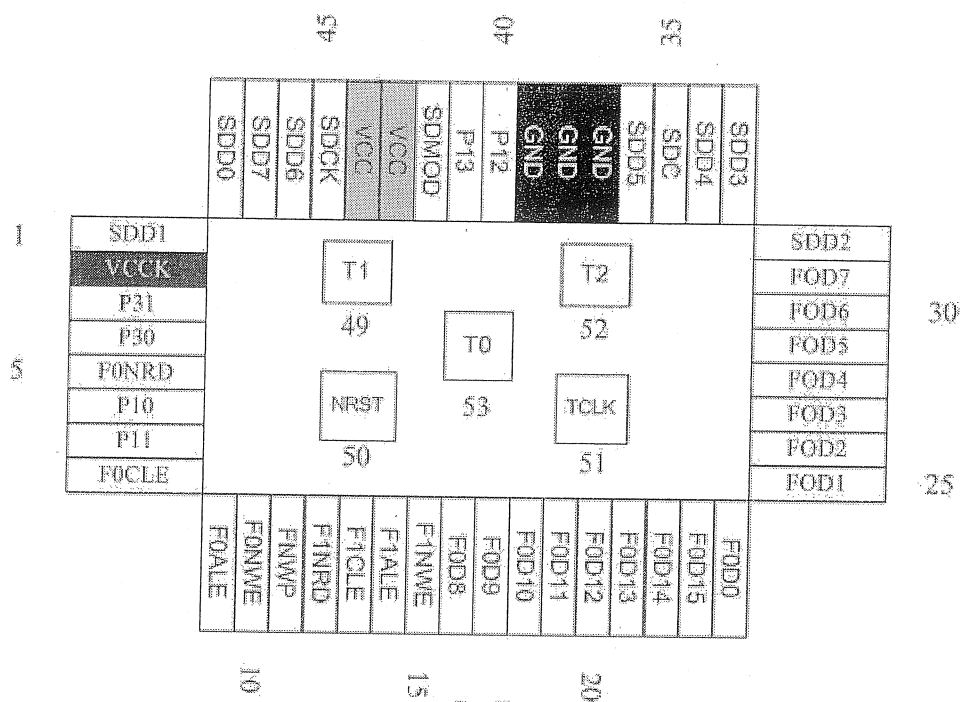


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Pad No.	Text	X-axis	Y-axis
73	F1NRD	33.0	950.0
74	P1[0]	33.0	874.0
75	P1[1]	33.0	798.0
76	GND	33.0	722.0
77	GND	33.0	646.0
78	P1[2]	33.0	570.0
79	P1[3]	33.0	494.0
80	GNDA	33.0	342.0
81	VCCA	33.0	266.0
82	V18	33.0	190.0
83	V18	33.0	114.0
84	VCCA	33.0	38.0

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**Physical Pin Assignment Data Part III : 48 pin LGA package**
(7 mm x 3.8 mm x 0.7 mm)**Figure 6 Physical Pin Assignment Data Part III : 48 pin LGA package**

No.	Name	No.	Name	No.	Name
1	SDD[1]	17	F0D[9]	33	SDD[3]
2	VCCCK	18	F0D[10]	34	SDD[4]
3	P3[1]	19	F0D[11]	35	SDC
4	P3[0]	20	F0D[12]	36	SDD[5]
5	FONRD	21	F0D[13]	37	GND
6	P1[0]	22	F0D[14]	38	GND
7	P1[1]	23	F0D[15]	39	GND
8	FOCLE	24	F0D[0]	40	P1[2]
9	F0ALE	25	F0D[1]	41	P1[3]
10	F0NWE	26	F0D[2]	42	SDMOD
11	FNWP	27	F0D[3]	43	VCC
12	F1NRD	28	F0D[4]	44	VCC
13	F1CLE	29	F0D[5]	45	SDCK
14	F1ALE	30	F0D[6]	46	SDD[6]
15	F1NWE	31	F0D[7]	47	SDD[7]
16	F0D[8]	32	SDD[2]	48	SDD[0]



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48 pin 7mm x 3.8mm LGA package mechanical drawing

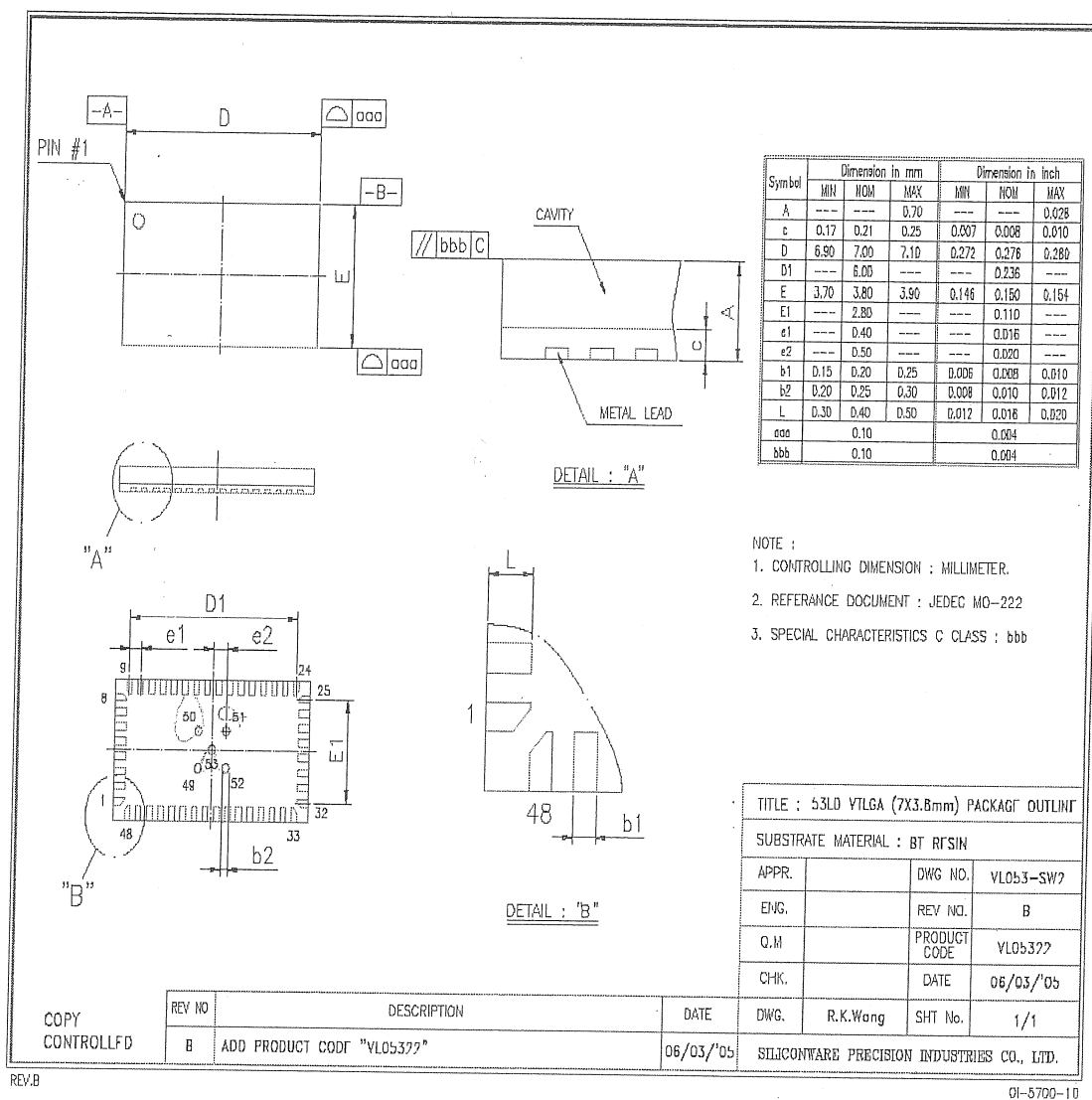


Figure 7 48 pin 7mm x 3.8mm LGA package mechanical drawing



Chapter 3: Register Table

⇒ 253/1/1/22 자 확인하기?

This section introduce the register in MMC and the values that SM265 used. The following table is the register list of current specification. The detail functionality is not described here, please reference to latest MMC specifications.

Register Name	MMC 4.0/4.1	MMC 3.3
Operation Condition Register (OCR)	V	V
Card Identification Register (CID)	V	V
Driver Stage Register (DSR)	V	V
Relative Card Address Register (RCA)	V	V
Card Specific Data Register (CSD)	V	V
Extended CSD Register (EXT_CSD)	V	
SD card Configuration Register (SCR)		

Operation Condition Register (OCR)

The 32-bit operation conditions register stores the V_{DD} voltage profile of the card. In addition, this register includes a status information bit. This status bit is set if the card power up procedure has been finished. The OCR register shall be implemented by all cards. The supported voltage range is coded as shown in the following table, for Single voltage MMC. As long as the card is busy, the corresponding bit (31) is set to LOW.

OCR bit	VDD voltage window	Single voltage MMC
[6:0]	Reserved	000 0000 b
[7]	1.65V – 1.95V	0 b
[14:8]	2.0V – 2.6V	000 0000 b
[23:15]	2.7V – 3.6V	1 1111 1111 b
[30:24]	Reserved	000 0000 b
[31]	Card power up status bit	

(1) OCR bit [31] is set to LOW if the card has not finished the power up routine.

Card Identification Register (CID) - 자 확인하기

The Card Identification (CID) register is 128 bits wide. It contains the card identification information used during the card identification phase. Every individual flash or I/O card shall have an unique identification number. The structure of the CID register is defined in the following table.

CID bit	width	Name	Field
[127:120]	8	Manufacture ID	MID
[119:104]	16	OEM/Application ID	OID
[103:56]	48	Product Name	PNM
[55:48]	8	Product Revision	PRV
[47:16]	32	Product Serial Number	PSN
[15:8]	8	Manufacturing Date	MDT



[7:1]	7	CRC7 check sum	CRC
[0]	1	Not used, always '1'	---

- (1) All content in the CID table is programmable. Manufacturer can update the CID data through SMI utility.
(2) Manufacturer should license MID & OID field from MMCA.

Driver Stage Register (DSR)

The 16-bit driver stage register is optionally used to improve the bus performance for extended operating conditions. The CSD register carries the information about the DSR register usage. This register is not implemented in SM265. It is optional.

Relative Card Address Register (RCA)

The writable 16-bit relative card address register carries the card address assigned by the host during the card identification. This address is used for the addressed host-card communication after the card identification procedure. The default value of the RCA register is 0x0001. The value 0x0000 is reserved to set all cards into the Stand-by State with CMD7.

In MMC mode, the value of this register is assigned by host.

Card Specific Data Register (CSD)

The Card-Specific Data register provides information on how to access the card contents. The CSD defines the data format, error correction type, maximum data access time, data transfer speed, whether the DSR register can be used etc. The programmable part of the register can be changed by CMD27.



MMC 4.0/4.1 CSD table

CSD bit	Width	Name	Field	Value	Note
[127:126]	2	CSD structure	CSD_STRUCTURE	11 b	Higher than v3.3
[125:122]	4	Specification version	SPEC_VERS	0100 b	V4.0/4.1
[121:120]	2	Reserved	---	---	---
[119:112]	8	Data read access-time 1	TAAC	5E h	
[111:104]	8	Data read access-time 2	NSAC	00 h	
[103:96]	8	Max. bus clock freq.	TRAN_SPEED	2A h	20 MHz (*1)
[95:84]	12	Card command classes	CCC	1F5 h	(*2)
[83:80]	4	Max. read data block length	READ_BL_LEN	9 h	512 bytes
[79]	1	Partial block read allowed	READ_BL_PARTIAL	1 b	Support
[78]	1	Write block misalignment	WRITE_BLK_MISALIGN	0 b	Support
[77]	1	Read block misalignment	READ_BLK_MISALIGN	0 b	Support
[76]	1	DSR implemented	DSR_IMP	0 b	Not support
[75:74]	2	Reserved	---	---	---
[73:62]	12	Device size	C_SIZE	(*3)	(*3)
[61:59]	3	Max. R_curr @ V _{DD} min	VDD_R_CURR_MIN	101 b	35 mA
[58:56]	3	Max R_curr @ V _{DD} max	VDD_R_CURR_MAX	101 b	45 mA
[55:53]	3	Max W_curr @ V _{DD} min	VDD_W_CURR_MIN	101 b	35 mA
[52:50]	3	Max W_curr @ V _{DD} max	VDD_W_CURR_MAX	101 b	45 mA
[49:47]	3	Device size multiplier	C_SIZE_MULT	(*3)	(*3)
[46:42]	5	Erase group size	ERASE_GRP_SIZE	(*4)	(*4)
[41:37]	5	Erase group size multiplier	ERASE_GRP_SIZE_MULT	(*4)	(*4)
[36:32]	5	Write protect group size	WP_GRP_SIZE	(*5)	(*5)
[31]	1	Write protect group enable	WP_GRP_ENABLE	1 b	Support
[30:29]	2	Manufacturer default ECC	DEFAULT_ECC	00 b	None
[28:26]	3	Write speed factor	R2W_FACTOR	101 b	32X
[25:22]	4	Max. write data block length	WRITE_BL_LEN	9 h	512 bytes
[21]	1	Partial block write allowed	WRITE_BL_PARTIAL	0 b	Not Support
[20:17]	4	Reserved	---	---	---
[16]	1	Content protection application	CONTENT_PROT_APP	0 b	Not support
[15]	1	File format group	FILE_FORMAT_GRP	0 b	HD like FAT
[14]	1	Copy flag	COPY	0 b	Not copied
[13]	1	Permanent write protection	PERM_WRITE_PROTECT	0 b	Not protected
[12]	1	Temporary write protection	TMP_WRITE_PROTECT	0 b	Not protected
[11:10]	2	File format	FILE_FORMAT	00 b	HD like FAT
[9:8]	2	ECC code	ECC	00 b	None
[7:1]	7	CRC	CRC	---	---
[0]	1	Not used, always '1'	---	1 b	---

(*1) This field only index low speed mode data. Please reference EXT_CSD for high speed data.

(*2) Support command class 0,2,4,5,6,7. Include : Basic, Block read/write, Erase, Write protection, and Lock card.
Not support 1,3,8. Include : Stream read/write and application command.

(*3)~(*5) This field is not a constant value. The value will be changed by different flash memory. For example, the value of Samsung SLC flash is different from Toshiba MLC flash.

Note : bit [15:0] is programmable by host side. Please reference to MMC specification for detail information



MMC 3.3 CSD table

CSD bit	Width	Name	Field	Value	Note
[127:126]	2	CSD structure	CSD_STRUCTURE	10 b	v3.1 – v3.2
[125:122]	4	Specification version	SPEC_VERS	0011 b	v3.1 – v3.3
[121:120]	2	Reserved	---	---	---
[119:112]	8	Data read access-time 1	TAAC	5E h	
[111:104]	8	Data read access-time 2	NSAC	00 h	
[103:96]	8	Max. bus clock freq.	TRAN_SPEED	2A h	20 MHz
[95:84]	12	Card command classes	CCC	1F5 h	(*1)
[83:80]	4	Max. read data block length	READ_BL_LEN	9 h	512 bytes
[79]	1	Partial block read allowed	READ_BL_PARTIAL	1 b	Support
[78]	1	Write block misalignment	WRITE_BLK_MISALIGN	0 b	Support
[77]	1	Read block misalignment	READ_BLK_MISALIGN	0b	Support
[76]	1	DSR implemented	DSR_IMP	0 b	Not support
[75:74]	2	Reserved	---	---	---
[73:62]	12	Device size	C_SIZE	(*2)	(*2)
[61:59]	3	Max. R_curr @ V _{DD} min	VDD_R_CURR_MIN	101 b	35 mA
[58:56]	3	Max R_curr @ V _{DD} max	VDD_R_CURR_MAX	101 b	45 mA
[55:53]	3	Max W_curr @ V _{DD} min	VDD_W_CURR_MIN	101 b	35 mA
[52:50]	3	Max W_curr @ V _{DD} max	VDD_W_CURR_MAX	101 b	45 mA
[49:47]	3	Device size multiplier	C_SIZE_MULT	(*2)	(*2)
[46:42]	5	Erase group size	ERASE_GRP_SIZE	(*3)	(*3)
[41:37]	5	Erase group size multiplier	ERASE_GRP_SIZE_MULT	(*3)	(*3)
[36:32]	5	Write protect group size	WP_GRP_SIZE	(*4)	(*4)
[31]	1	Write protect group enable	WP_GRP_ENABLE	1 b	Support
[30:29]	2	Manufacturer default ECC	DEFAULT_ECC	00 b	None
[28:26]	3	Write speed factor	R2W_FACTOR	101 b	32X
[25:22]	4	Max. write data block length	WRITE_BL_LEN	9 h	512 bytes
[21]	1	Partial block write allowed	WRITE_BL_PARTIAL	0 b	Not Support
[20:17]	4	Reserved	---	---	---
[16]	1	Content protection application	CONTENT_PROT_APP	0 b	Not support
[15]	1	File format group	FILE_FORMAT_GRP	0 b	HD like FAT
[14]	1	Copy flag	COPY	0 b	Not copied
[13]	1	Permanent write protection	PERM_WRITE_PROTECT	0 b	Not protected
[12]	1	Temporary write protection	TMP_WRITE_PROTECT	0 b	Not protected
[11:10]	2	File format	FILE_FORMAT	00 b	HD like FAT
[9:8]	2	ECC code	ECC	00 b	None
[7:1]	7	CRC	CRC	---	---
[0]	1	Not used, always '1'	---	1 b	---

(*1) Support command class 0,2,4,5,6,7. Include : Basic, Block read/write, Erase, Write protection, and Lock card.
Not support 1,3,8. Include : Stream read/write and application command.

(*2)~(*4) This field is not a constant value. The value will be changed by different flash memory. For example, the value of Samsung SLC flash is different from Toshiba MLC flash.

Note : bit [15:0] is programmable by host side. Please reference to SD/MMC specification for detail information.

**Extended CSD Register (EXT_CSD)**

The Extended CSD register defines the card properties and selected modes. It is 512 bytes long.

The most significant 320 bytes are the Properties segment, which defines the card capabilities and cannot be modified by the host. The lower 192 bytes are the Modes segment, which defines the configuration the card is working in. These modes can be changed by the host by means of the SWITCH command.

For MMC 4.0/4.1 only

EXT_CSD (bytes)	Width (bytes)	Name	Field	Value	Note
Properties segment					
[511:505]	7	Reserved	---	---	---
[504]	1	Supported command sets	S_CMD_SET (R)	00 h	Standard MMC
[503:204]	300	Reserved	---	---	---
[203]	1	Power class for 26MHz @ 3.6V	PWR_CL_26_360 (R)	(*1)	(*1)
[202]	1	Power class for 52MHz @ 3.6V	PWR_CL_52_360 (R)	(*1)	(*1)
[201]	1	Power class for 26MHz @ 1.95V	PWR_CL_26_195 (R)	(*1)	(*1)
[200]	1	Power class for 52MHz @ 1.95V	PWR_CL_52_195 (R)	(*1)	(*1)
[199:197]	3	Reserved	---	---	---
[196]	1	Card type	CARD_TYPE (R)	03 h	HS-MMC @ 52MHz
[195]	1	Reserved	---	---	---
[194]	1	CSD structure version	CSD_STRUCTURE (R)	03 h	v4.0/4.1
[193]	1	Reserved	---	---	---
[192]	1	Extension CSD Reversion	EXT_CSD_REV (R)	00 h	Revision 1.0
Modes segment					
[191]	1	Command set	CMD_SET (R/W)	Setup by host	
[190]	1	Reserved	---	---	---
[189]	1	Command set revision	CMD_SET_REV (RO)	00 h	v4.0
[188]	1	Reserved	---	---	---
[187]	1	Power class	POWER_CLASS (R/W)	Setup by host	
[186]	1	Reserved	---	---	---
[185]	1	High speed interface timing	HS_TIMING (R/W)	Setup by host	
[184]	1	Reserved	---	---	---
[183]	1	Bus width mode	BUS_WIDTH (WO)	Setup by host	
[182:0]	183	Reserved	---	---	---

(*1) Not finalized.



Chapter 4: AC/DC character

General DC Character

Parameter	Symbol	Min.	Max.	Unit	Remark
Peak voltage on all lines		-0.5	3.6	V	
All input leakage current		-1	1	uA	Without pull up R
All output leakage current		-1	1	uA	Without pull up R
Supply voltage for low voltage range	V_{DDL}	1.65	1.95	V	
Supply voltage for high voltage range	V_{DDH}	2.7	3.6	V	
Supply voltage differential		-0.5	0.5	V	

Bus Signal Line Loading

Parameter	Symbol	Min.	Max.	Unit	Remark
Pull up resistance for SDC line	R_{CMD}	4.7	100	K Ohm	To prevent bus floating
Pull up resistance for SDD line	R_{DAT}	50	100	K Ohm	To prevent bus floating
Bus line capacitance	C_L		30	pF	Single card
Signal line inductance			16	nH	

Bus Signal Level

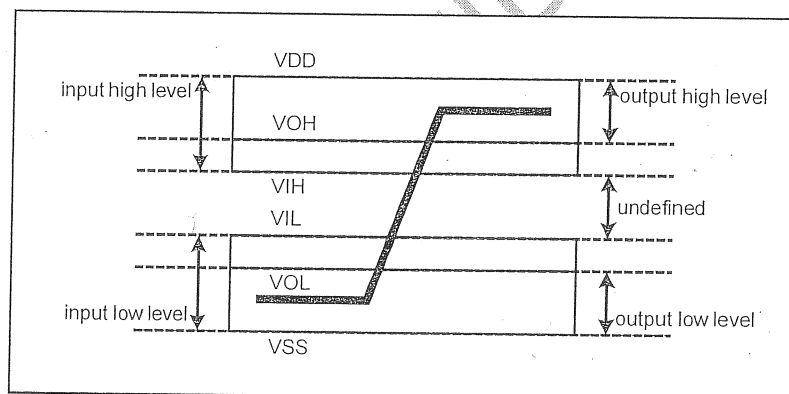


Figure 8: Bus Signal Level

Parameter	Symbol	Min.	Max.	Unit	Remark
Output High voltage	V_{OH}	2.4		V	$V_{DD} = 3.3V$
Output Low voltage	V_{OL}		0.4	V	$V_{DD} = 3.3V$
Input High voltage	V_{IH}	2.0		V	$V_{DD} = 3.3V$
Input Low voltage	V_{IL}		0.8	V	$V_{DD} = 3.3V$



Bus Timing

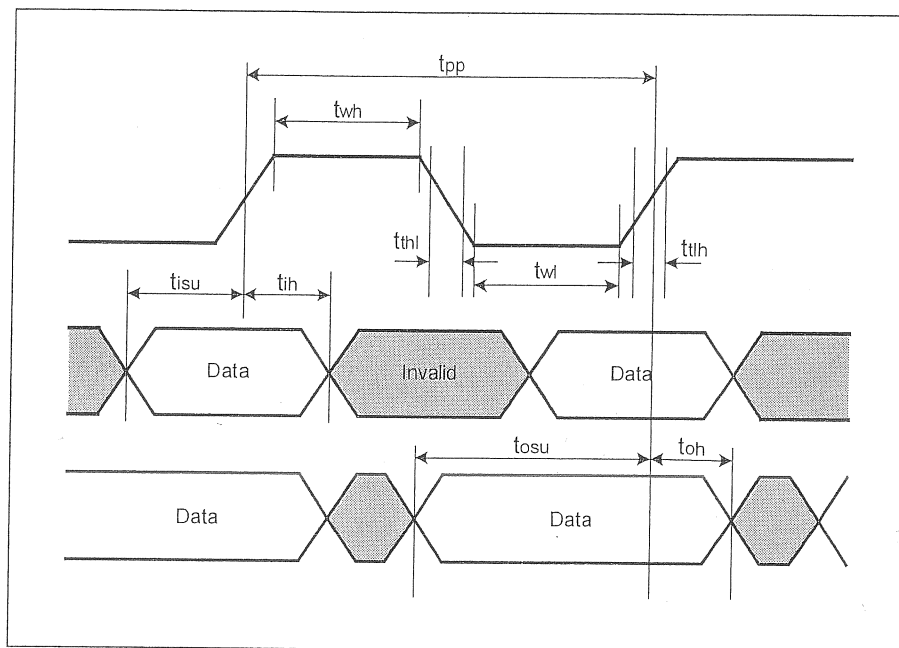


Figure 9: Bus Timing

High Speed Mode Timing

Parameter	Symbol	Min.	Max.	Unit	Remark
Input SDCK					
Clock frequency data transfer mode	f_{pp}	26	52	MHz	$C_L \leq 30\text{pF}$
Clock low time / Clock high time	t_{WL}/t_{WH}	6.5		ns	$C_L \leq 30\text{pF}$
Clock rise time / Clock fall time	t_{TLH}/t_{THL}		3	ns	$C_L \leq 30\text{pF}$
Input SDD/SDC, reference to SDCK					
Input set-up time	t_{ISU}	3		ns	$C_L \leq 30\text{pF}$
Input hold time	t_{IH}	3		ns	$C_L \leq 30\text{pF}$
Output SDD/SDC, reference to SDCK					
Output set-up time	t_{OSU}	5		ns	$C_L \leq 30\text{pF}$
Output hold time	t_{OH}	6		ns	$C_L \leq 30\text{pF}$
Signal rise time	t_{RISE}		2.5	ns	$C_L \leq 30\text{pF}$
Signal fall time	t_{FALL}		2.5	ns	$C_L \leq 30\text{pF}$

- (1) All timing values are measured relative to 50% of voltage level.
(2) Rise and fall times are measured from 10% - 90% of voltage level.

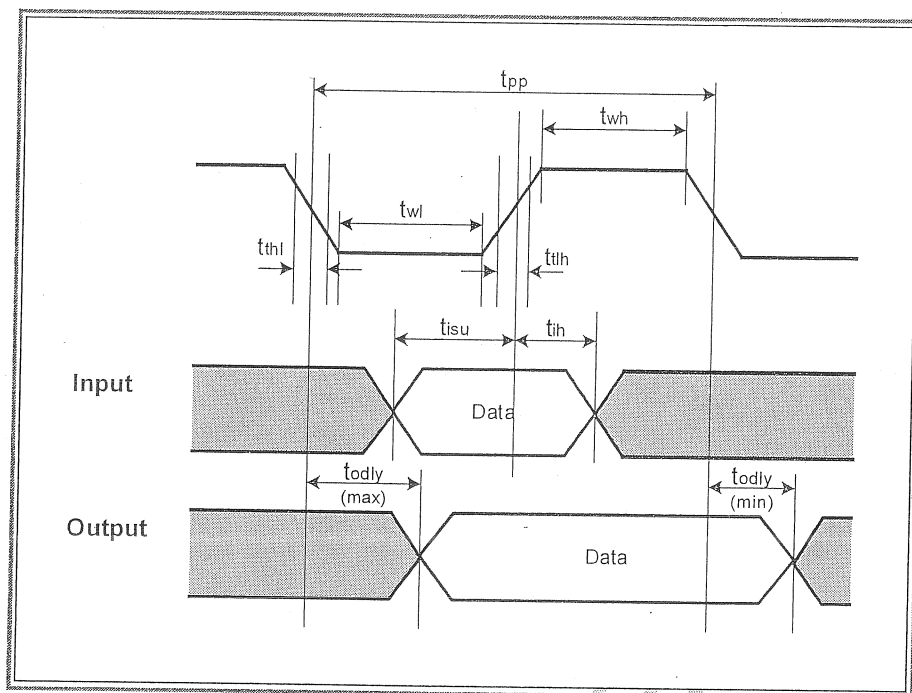


Figure 10: High Speed Mode Timing

Low Speed Mode Timing (Backward compatible)

Parameter	Symbol	Min.	Max.	Unit	Remark
Input SDCK					
Clock frequency data transfer mode	f_{pp}	0	25	MHz	$C_L \leq 30pF$
Clock frequency identification mode	f_{od}	0	400	KHz	$C_L \leq 30pF$
Clock low time / Clock high time	t_{WL}/t_{WH}	10		ns	$C_L \leq 30pF$
Clock rise time / Clock fall time	t_{TLH}/t_{THL}		10	ns	$C_L \leq 30pF$
Input SDD/SDC, reference to SDCK					
Input set-up time	t_{ISU}	3		ns	$C_L \leq 30pF$
Input hold time	t_{IH}	3		ns	$C_L \leq 30pF$
Output SDD/SDC, reference to SDCK					
Output delay time	t_{Ody}	3	7	ns	$C_L \leq 30pF$

- (1) All timing values are measured relative to 50% of voltage level.
 (2) Clock Rise and fall times are measured from V_{IH} to V_{IL} of voltage level.



Flash Interface AC Characteristics

Parameter	Symbol	Min.	Max.	Unit	Remark
CLE Setup Time	tCLS	27.5		ns	
CLE Hold Time	tCLH	12.5		ns	
ALE Setup Time	tALS	25		ns	
ALE Hold Time	tALH	10		ns	
WE Pulse Width	tWP	15		ns	
Data Setup Time	tDS	25		ns	
Data Hold Time	tDH	12.5		ns	
Write Cycle Time	tWC	37.5		ns	
WE High Hold Time	tWH	20		ns	
Read Cycle Time	tRC	37.5		ns	
RE pulse Width	tRP	17.5		ns	
RE High Hold Time	tREH	20		ns	

Command Latch Cycle

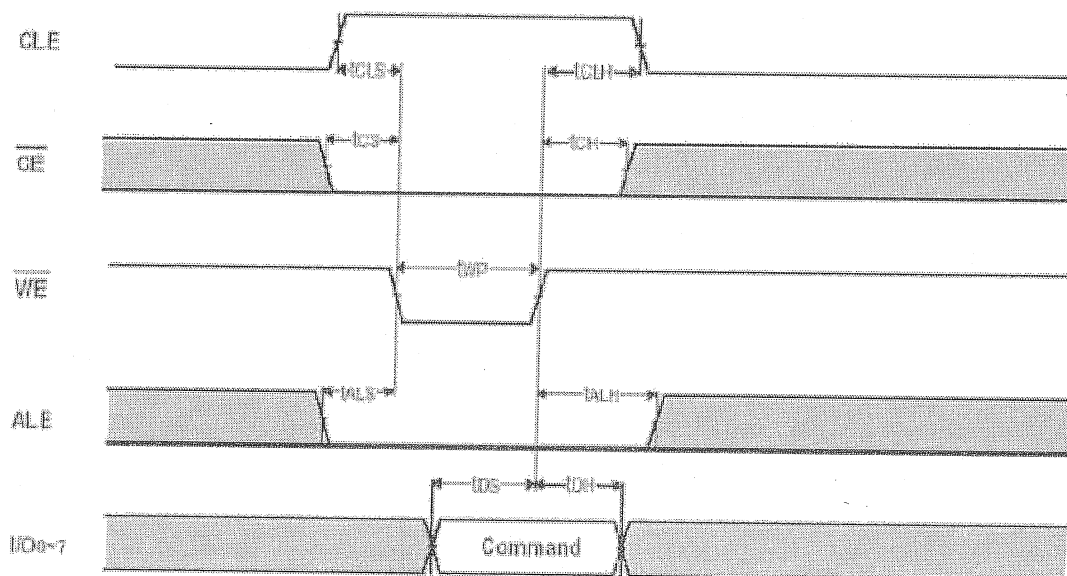


Figure 9: Command Latch Cycle



Sequential Out Cycle after Read (CLE=L, /WE=H, ALE=L)

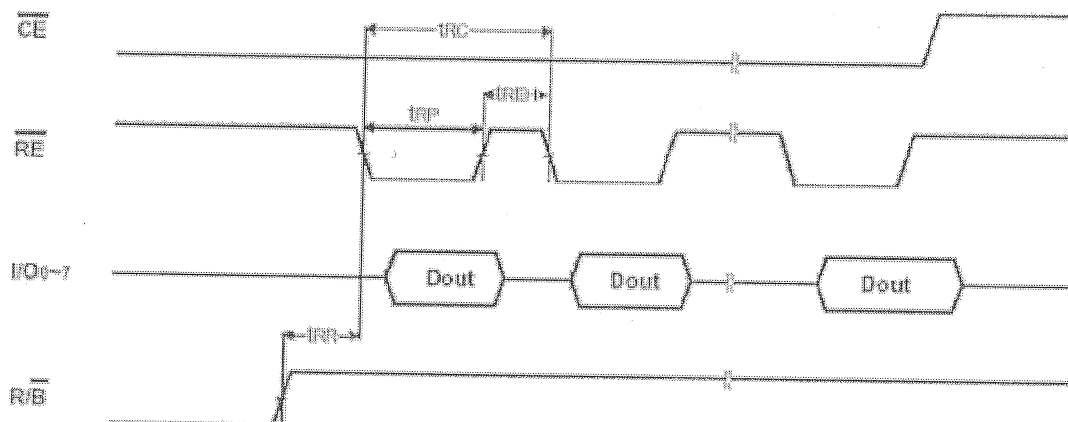


Figure 12: Sequential Out Cycle after Read

Power Consumption

Parameter	Symbol	Typ.	Max.	Unit	Remark
Read current at 1.8V			25	mA	$V_{DD} = 1.8V$
Write current at 1.8V			25	mA	$V_{DD} = 1.8V$
Pre-initialization standby current at 1.8V				uA	$V_{DD} = 1.8V$
Post-initialization standby current at 1.8V				uA	$V_{DD} = 1.8V$
Read current at 3.3V			19	mA	$V_{DD} = 3.3V$
Write current at 3.3V			19	mA	$V_{DD} = 3.3V$
Pre-initialization standby current at 3.3V		130	150	uA	$V_{DD} = 3.3V$
Post-initialization standby current at 3.3V		130	150	uA	$V_{DD} = 3.3V$

Note: All power consumption value are measured with one Samsung 1G bits flash (K9F1G08U0M).



Chapter 5: Reliability and Durability

Temperature	Operation : -25°C to 85°C
ESD protection	Contact Pad : (1) More than +/- 5KV, Human Body Model according to ANSI EOS/ESD-S5.1-1998 (Charge C = 100pF, Discharge R = 1.5 K Ohm) (2) More than +/- 5KV, Human Body Model according to IEC61000-4-2 (Charge C = 100pF, Discharge R = 1.5 K Ohm)

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